

FINAL REPORT

Team ID: LTVIP2026TMIDS47882

Title: Heart Disease Analysis

1.INTRODUCTION:

1.1 Project Overview

Heart disease remains one of the leading causes of mortality worldwide. Its prevalence continues to rise due to lifestyle changes, unhealthy dietary habits, physical inactivity, and increasing stress levels. Despite advancements in medical science, prevention and early detection remain critical in reducing severe outcomes and improving patient survival rates. Analysing large-scale healthcare data—including patient demographics, medical history, lifestyle choices, and clinical indicators—requires advanced data visualization tools to extract meaningful insights.

1.2 Purpose

The purpose of this project is to analyse heart disease data using Tableau to identify major risk factors affecting cardiovascular health. The project aims to transform raw healthcare data into meaningful and interactive visualizations. It focuses on understanding how age, BMI, lifestyle habits, and medical conditions influence heart disease occurrence. By uncovering patterns and correlations, the analysis supports data-driven decision-making. Ultimately, the project promotes awareness and preventive strategies to reduce heart disease risk.

2. IDEATION PHASE

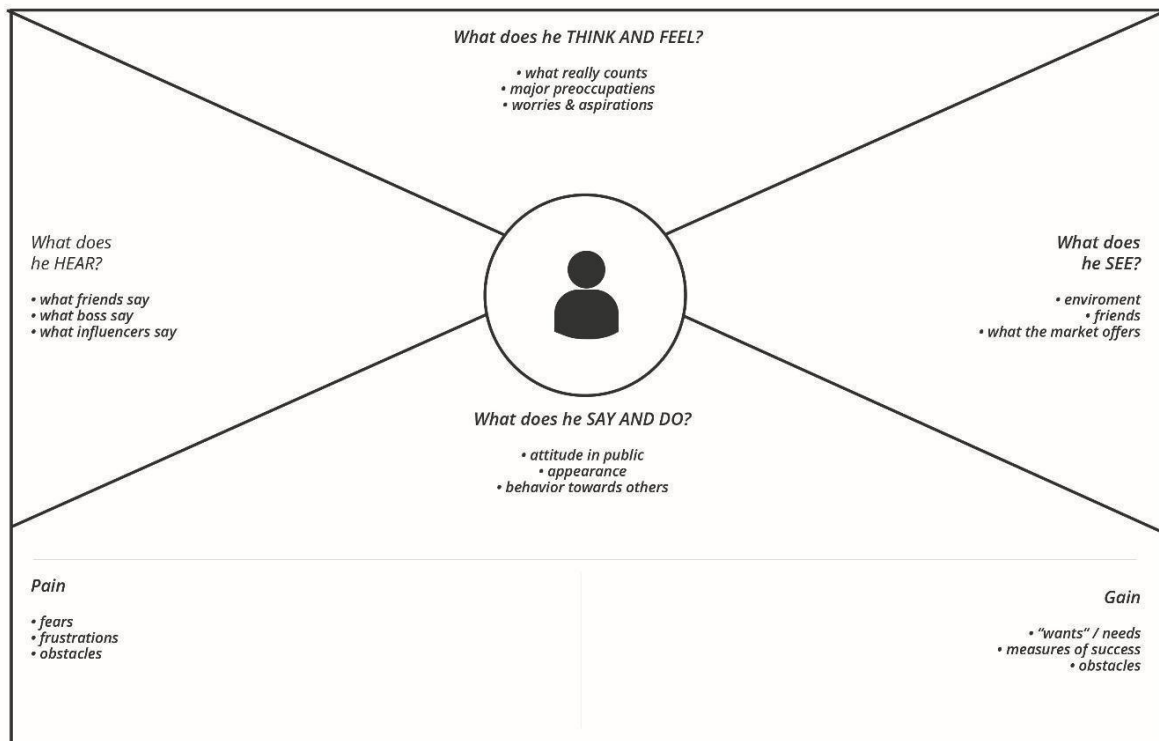
2.1 Problem Statement

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS – 1	A doctor or healthcare professional handling heart disease patients.	Analyze patient health data quickly to identify heart disease risks early.	Manual analysis of multiple medical parameters takes time and may lead to error	Large volumes of patient data make it difficult to identify important risk factors accurately.	Overwhelmed and concerned about missing early warning signs in patients.
PS – 2	A patient	Understand	Medical	There is no	Confused and

	concerned about my heart health.	my risk level and take preventive action early	reports are complex and difficult for me to interpret.	simple system that clearly explains heart disease risk in an easy way.	anxious about my health condition.
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2.2 Empathy Map Canvas

Empathy Map



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Business Model **Toolbox**

2.3 Brainstorming

S. No	Ideas Generated	Category
1	Visualize heart disease distribution across different age groups	Age-based Analysis
2	Compare heart disease occurrence between male and female patients	Gender Comparison
3	Analyze BMI distribution and its relationship with heart disease	Health Risk Assessment
4	Study the impact of smoking and alcohol consumption on heart disease	Lifestyle Analysis
5	Examine the relationship between diabetes and heart disease	Medical Condition Analysis
6	Analyze stroke history and its influence on heart disease occurrence	Risk Factor Evaluation
7	Compare heart disease rates among physically active vs inactive individuals	Behavioral Analysis
8	Highlight states with highest and lowest per capita electricity usage	Health Status Comparison
9	Show changes in urban vs rural electricity consumption	Visualization Enhancements
10	Integrate external datasets like population or weather for deeper insights	Insight & Decision Support

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Customer Journey Stages:

Stage 1: Problem Awareness

User Realization:

- Heart disease cases are rising due to lifestyle factors.
- Lack of early detection tools.

- Difficulty in comparing patient risk profiles.

Pain Points:

- Raw health data is hard to interpret.
- No centralized dashboard for risk analysis.
- Limited visibility into regional and demographic trends.

Stage 2: Data Exploration

User Actions:

- Access Tableau dashboard
- Filter by age, gender, region, and lifestyle indicators.
- Compare patient groups and risk factors.

System Response:

- Displays interactive graphs and heatmaps
- Highlights top risk contributors (e.g., smoking, high BMI)
- Enables regional and demographic comparisons.

Stage 3: Insight Discovery

User Discovers:

- Middle-aged patients with high cholesterol and smoking habits are at greatest risk
- Urban regions show higher prevalence due to sedentary lifestyle
- Lockdown periods reduced physical activity, increasing risk.
- Certain demographics show consistent upward trends in heart disease.

Stage 4: Decision Making

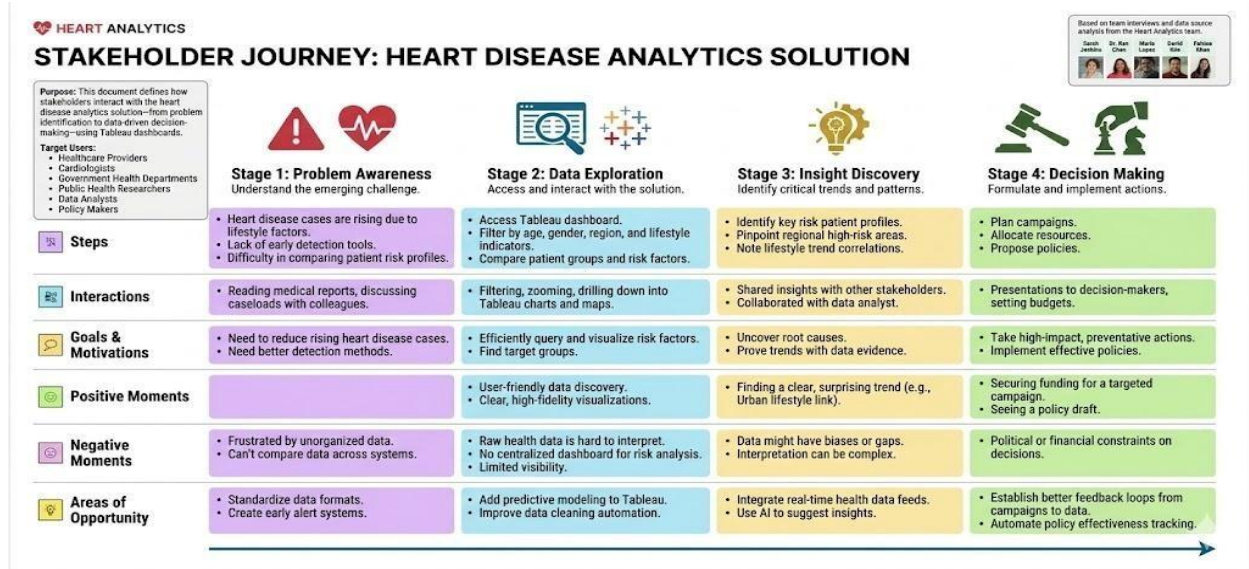
User Decisions:

- Launched targeted awareness campaigns for high-risk groups
- Promote lifestyle changes like physical activity and healthy diets
- Allocate resources for early screening in vulnerable regions

- Develop preventive health policies and incentives

Customer Journey Visualization Flow

- Problem + Data Access + Filtering → Visualization → Insights → Strategic Decision



3.2 Solution Requirement

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form, Registration through Gmail, Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email, Confirmation via OTP
FR-3	User Login & Authentication	Login using Email & Password, Forgot Password, Session Management
FR-4	Data Upload & Management	Upload patient health data, Validate input data, Store data securely
FR-5	Data Preprocessing	Handle missing values, Feature scaling, Data normalization

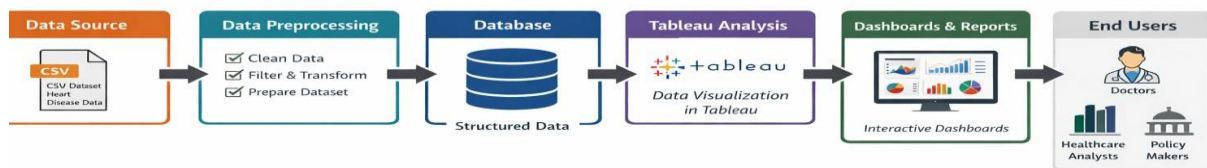
Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should be easy to use with a simple and intuitive interface for users and doctors.
NFR-2	Security	User data and medical information must be securely stored with authentication and authorization mechanisms.
NFR-3	Reliability	The system should provide accurate and consistent prediction results with minimal failures.
NFR-4	Performance	The system should process data and display predictions quickly with minimal response time.
NFR-5	Availability	The application should be accessible whenever required with minimal downtime.
NFR-6	Scalability	The system should support increased users and larger datasets without performance degradation.

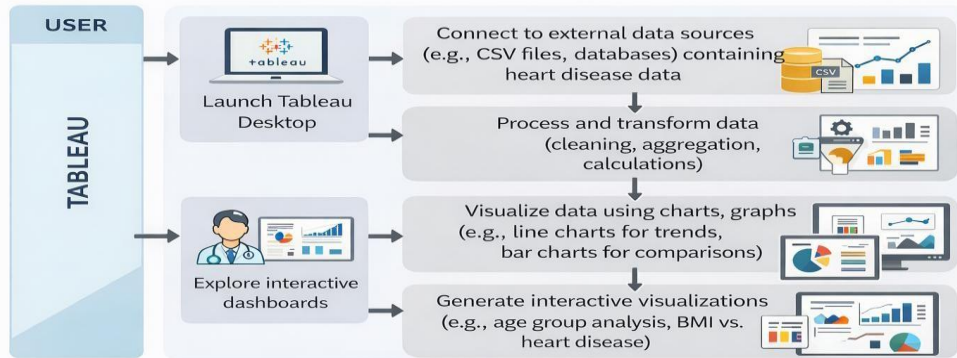
3.3 Data Flow Diagram

Heart Disease Analysis — Data Flow Diagram



3.4 Technology Stack

Tableau Workflow for Heart Disease Analysis



4. PROJECT DESIGN

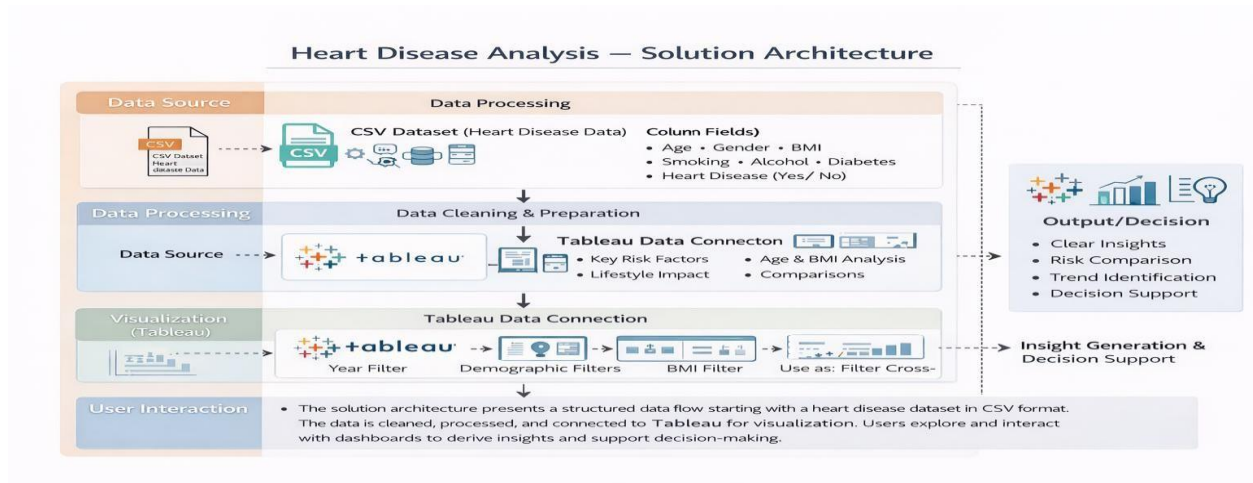
4.1 Problem Solution Fit

1. CUSTOMER SEGMENT(S) C5 Energy analysts and government planners analyzing electricity consumption data	6. CUSTOMER CONSTRAINTS CC Electricity consumption data is stored in large, raw Excel sheets making interpretation very difficult. Manually comparing year-by-year, state-by-state, and region-by-region consumption data is time analysis.	5. AVAILABLE SOLUTIONS A5 Currently, the data is in large Excel sheets and difficult to compare. Simple charts in Excel are static, and provide interactive features required for deep analysis.
3. JOBS-TO-BE-DONE / PROBLEMS J&P 1. Interpreting raw Excel data is challenging. 2. Comparing electricity consumption year-wise and region-wise is difficult. 3. Identifying Top N and Bottom N consuming states is hard. 4. Manual analysis is time-consuming and inefficient.	6. PROBLEM ROOT CAUSE RC Data is stored in large, unprocessed Excel sheets and lack of proper visualization tools.	7. BEHAVIOUR BE They manually sift through Excel sheets to compare yearly, regional, and state consumption data. Inefficient, and time-consuming decision-making.
3. TRIGGERS TR • When tasked to analyze electricity data, energy analysts manually analyze Excel sheets and compare states' consumption data.	1. YOUR SOLUTION SL Developed interactive Tableau dashboard for clear and meaningful insights with: • Year-wise bar charts comparing 2019 & 2020 • Region-wise distribution pie chart • Dynamic state-ranking (Top N & Bottom N) using parameters • Monthly trend analysis using line charts • Interactive filters (Year, Region, State) & "Use as Filter" feature.	6. BEHAVIOUR BE • They manually sift through Excel sheets to compare yearly, regional, and state consumption data. Inefficient, and time-consuming decision-making.
4. EMOTIONS: BEFORE / AFTER EM Before: Difficult, time-consuming, indecisive. After: Clear, fast, and confident decision-making.	9. YOUR SOLUTION SL • Developed interactive Tableau dashboard for clear and meaningful insights with: • Year-wise bar charts comparing 2019 & 2020 • Dynamic state-ranking tool for efficient decision-making	7. YOUR SOLUTION SL • Business & Marketing channel (ie late en iterees) • Analyst collaboration platform (mamber interetor).
5. EMOTIONS: BEFORE / AFTER EM Before: Difficult, time-consuming, indecisive. After: Clear, fast, and confident decision-making.		8. OUTCOME O 1. Solution effectively addresses the problems identified. 2. Enhanced efficiency and decision-making speed with interactive visual analysis.

4.2 Proposed Solution

S .No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Electricity consumption data is stored in large Excel sheets, making it difficult to interpret, compare year wise and region-wise trends, and identify high and low consuming states. Manual analysis is time consuming and inefficient for decision-making.
2.	Idea / Solution description	Develop an interactive Tableau dashboard that visualizes electricity consumption data using bar charts, line charts, and pie charts. The dashboard includes year-wise comparison, region-wise distribution, Top N & Bottom N state analysis, monthly trend visualization, and interactive filters (Year, Region, State).
3.	Novelty / Uniqueness	The solution integrates dynamic filtering, parameter based Top N and Bottom N analysis, and cross-sheet interaction in a single dashboard. It transforms static Excel data into an interactive, user-friendly analytical system.
4.	Social Impact / Customer Satisfaction	The dashboard helps energy planners and analysts make faster and more accurate decisions. It improves transparency in electricity consumption patterns and supports better energy planning and resource allocation.
5.	Business Model (Revenue Model)	The solution can be adopted by government agencies, electricity boards, and energy companies. It can be implemented as a data analytics service or integrated into energy management systems for performance monitoring.
6.	Scalability of the Solution	The dashboard can be expanded to include additional years, renewable energy data, predictive analytics, and real-time data integration. It can also be adapted for other sectors such as water consumption or fuel usage analysis. predictive analytics, and real-time data integration. It can also be adapted for other sectors such as water consumption or fuel usage analysis.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection & Preparation	USN-1	As an analyst, I can import heart disease data into Tableau.	3	High	P. Tilak
Sprint-1	Data Cleaning	USN-2	As an analyst, I can clean and structure the dataset (Age, Gender, Cholesterol).	3	High	Mounika Roy
Sprint-2	Dashboard Creation	USN-3	As a user, I can view risk factors using bar and pie charts.	5	High	M. Sai Gowtham Reddy
Sprint-2	Trend Analysis	USN-4	As a user, I can analyze lifestyle trends using line charts.	3	Medium	P. Tilak
Sprint-3	Filter Integration	USN-5	As a user, I can filter data by Age, Gender, Region and Smoking habits.	3	High	Mounika Roy

Sprint-3	Top/Bottom Analysis	USN-6	As a user, I can view low risk and high risk patient groups.	3	Medium	Gurram Pavithra Reddy
Sprint-4	Story & Final	USN-7	As a user, I can view insights in story	4	High	K Haritha Reddy

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	6	6 Days	15 Jan 2026	20 Jan 2026	6	20 Jan 2026
Sprint-2	8	6 Days	21 Jan 2026	28 Jan 2026	8	28 Jan 2026
Sprint-3	6	6 Days	29 Jan 2026	3 Feb 2026	6	3 Feb 2026
Sprint-4	4	6 Days	4 Feb 2026	7 Feb 2026	4	7 Feb 2026

Velocity:

Total Story Points Completed = 24 Number of Sprints = 4

Average Velocity = $24 \div 4 = 6$ Story Points per Sprint

If sprint duration = 6 days

Velocity per day = $6 \div 6 = 1$ Story Point per Day **Burndown**

Chart:

A Burndown Chart was used to track the remaining work against time during each sprint. The chart shows the ideal work completion line and the actual progress line. The project maintained steady progress, and story points were completed within the planned sprint duration, indicating effective sprint planning and execution.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

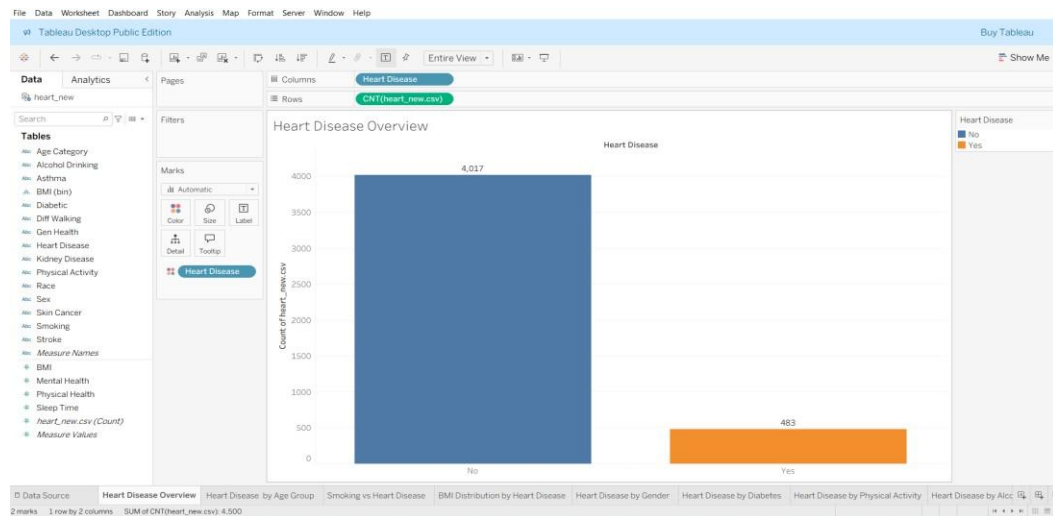
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	The dashboard successfully renders heart disease data without noticeable delay. The dataset includes Age Category, Gender, BMI, Smoking, Alcohol, Diabetes, Stroke, and heart disease fields. All visualizations load within acceptable response time.
2.	Data Preprocessing	Data was cleaned before visualization. Categorical fields were verified, BMI bins were created for histogram analysis, and null values were checked and handled appropriately.

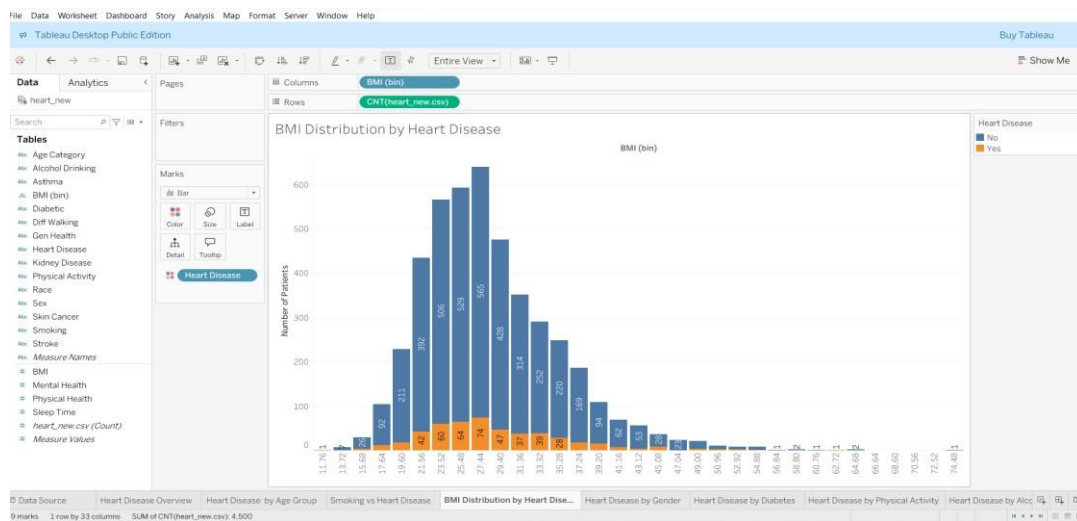
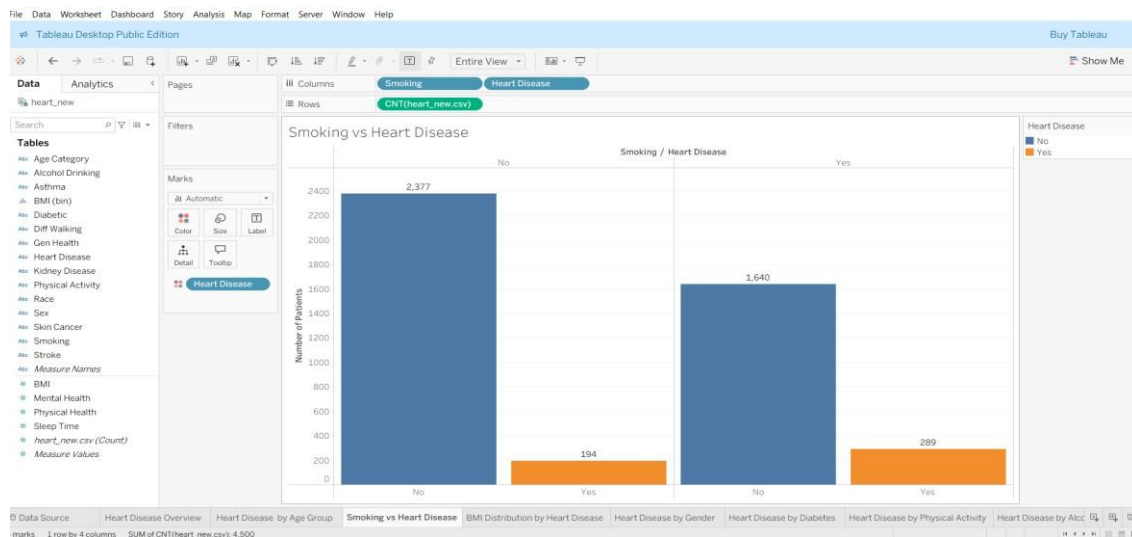
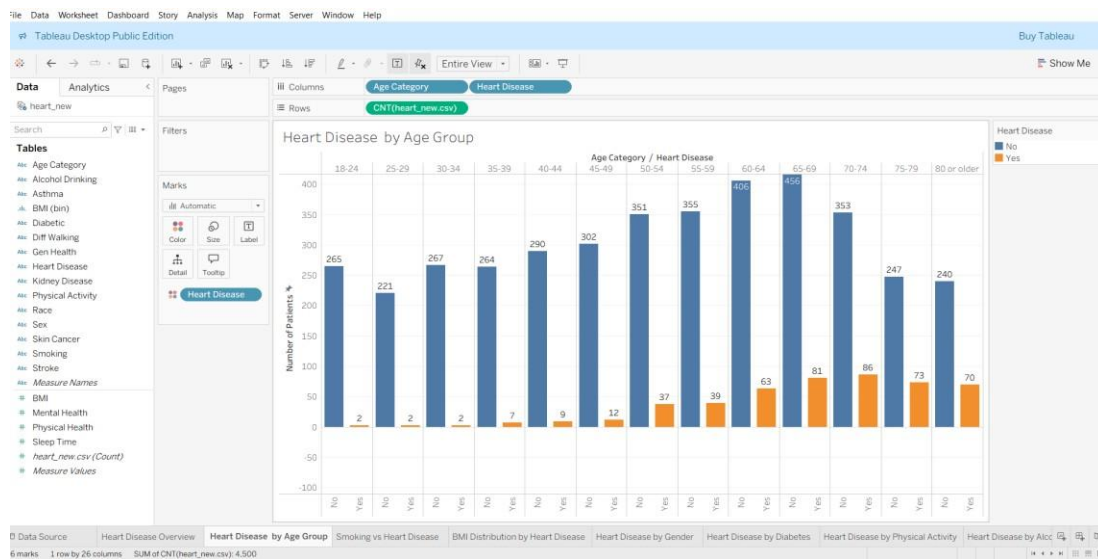
3.	Utilization of Filters	Interactive filters such as Age Category, Gender, BMI (bin), and Health Conditions were implemented. “Use as Filter” functionality enables cross-sheet interaction. Dashboards update dynamically when filters are applied.
4.	Calculation fields Used	Calculated fields such as COUNT of patients, BMI bin grouping, percentage calculation, and category-based aggregation were implemented to enhance analysis.
5.	Dashboard design	Number of Visualizations / Graphs – 10 (Bar Charts, Histogram, Comparative Analysis, Percentage Distribution, KPI Overview). Layout is structured, clean, and responsive.
6	Story Design	Story points are arranged logically to explain heart disease trends, risk factors, comparisons, and insights step by step. The storytelling approach improves understanding of the analysis.

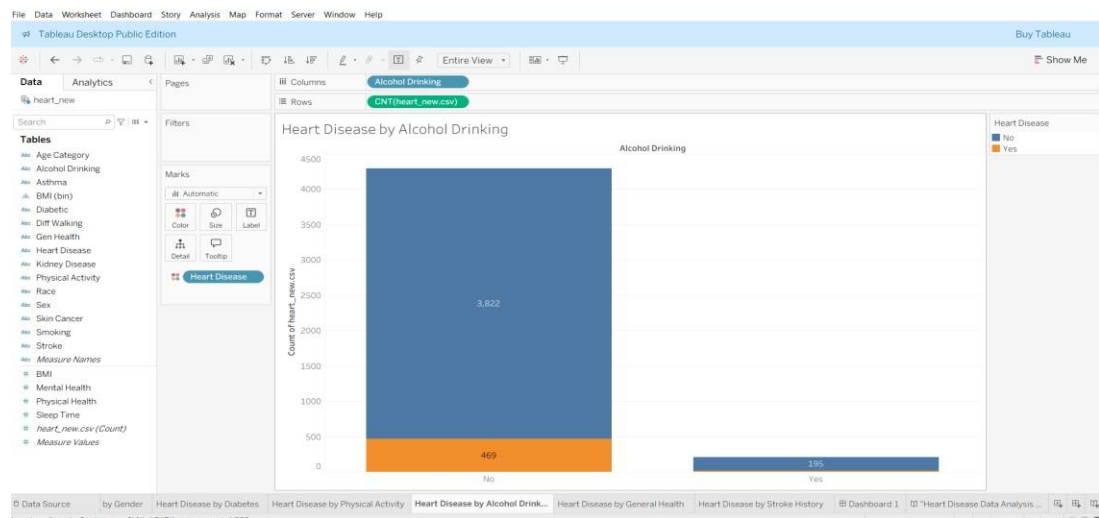
7. RESULTS

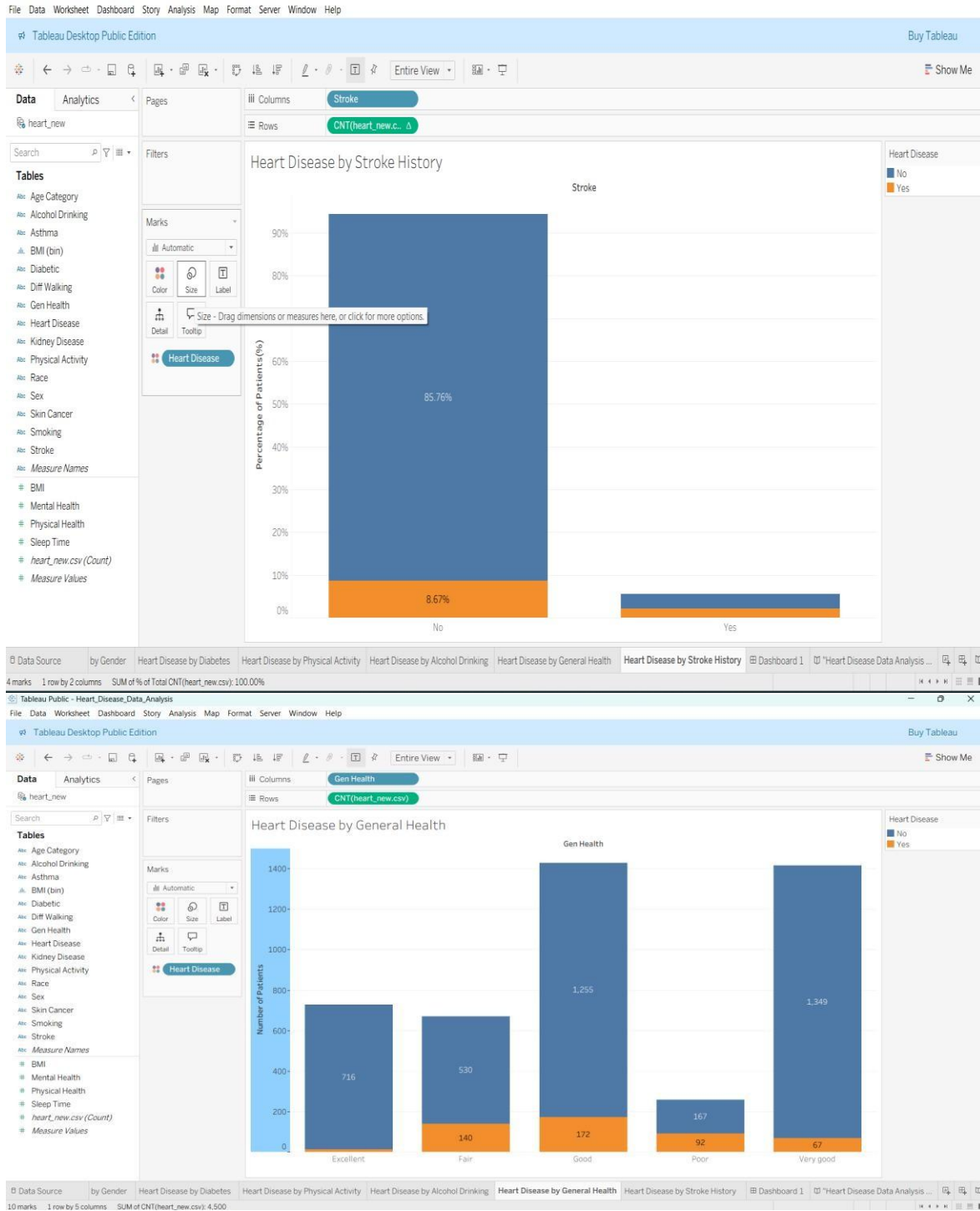
7.1 Output Screenshots

Dashboard Screenshots:





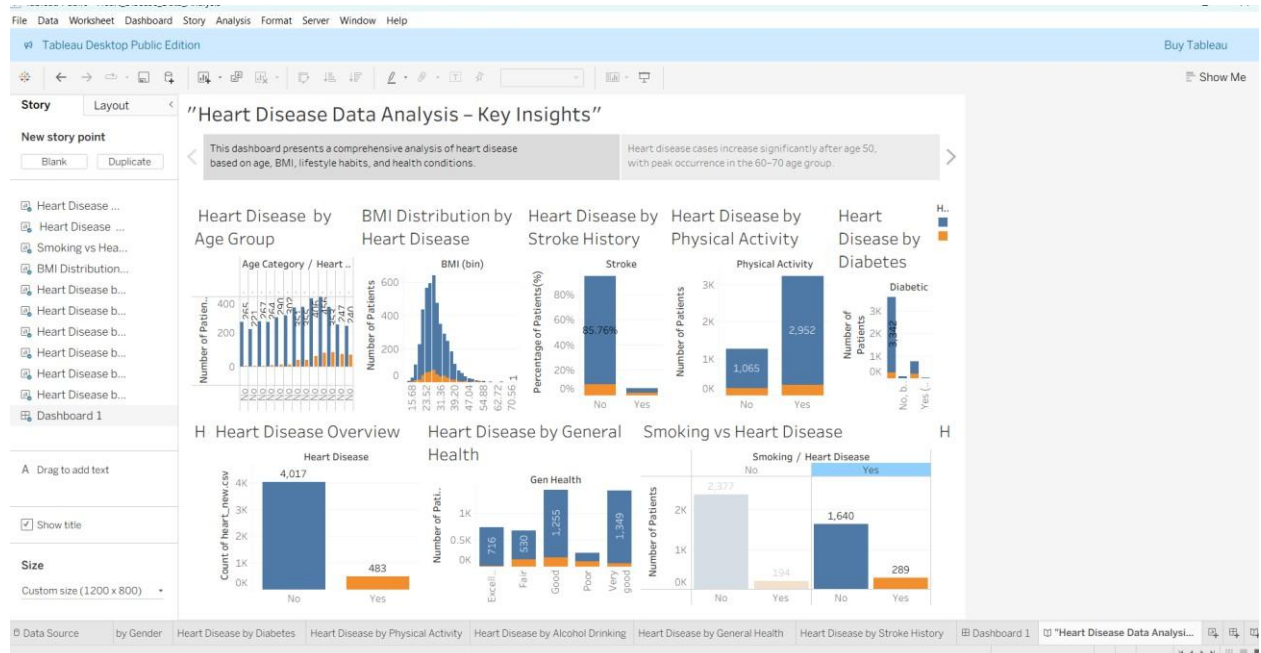




Dash Board



Story:



Web Integration



8. ADVANTAGES & DISADVANTAGES

Advantages

1. Interactive Visualization

The dashboard provides interactive charts and filters that allow users to explore electricity consumption data dynamically.

2. Improved Decision Making

Year-wise and region-wise analysis helps stakeholders make informed and data-driven decisions.

3. Easy Trend Identification

Monthly and seasonal trends can be easily identified through line charts and comparative analysis.

4. Time Saving

Instead of manually analyzing raw Excel data, the dashboard presents insights instantly.

5. User-Friendly Interface

The system is simple to use and does not require advanced technical knowledge.

6. Better Resource Planning

Helps energy planners identify high and low consumption regions for efficient energy distribution.

Disadvantages

- **Dependent on Data Quality**

If the input data contains errors or missing values, the analysis may not be accurate.

- **Limited to Available Data**

The dashboard can only analyze the data provided and cannot predict future consumption without advanced modeling.

- **Requires Tableau Software**

Users need access to Tableau to interact with the dashboard.

- **Initial Setup Time**

Data cleaning and dashboard development require initial time and effort.

9. CONCLUSION

This project analysed heart disease data using Tableau to identify key risk factors and trends. Interactive dashboards were created to study the impact of age, BMI, lifestyle habits, and medical conditions. The analysis revealed significant relationships between heart disease and factors such as diabetes, stroke, and physical inactivity. Data visualization helped transform raw healthcare data into meaningful insights. Overall, the project demonstrates the importance of data-driven decision-making in preventive healthcare.

10. FUTURE SCOPE

The project can be enhanced by integrating real-time healthcare datasets for continuous analysis. Machine learning models can be applied to predict heart disease risk more accurately. The dashboard can be deployed as a web application for wider accessibility. Advanced analytics such as trend forecasting can be incorporated. Additional health indicators can be included to improve risk assessment and decision support.

Git hub Link: <https://github.com/tilakworkspace19/Heart-Disease-Analyser.git>